

Risk-Aware Multi-Task Graph Neural Surrogate for Operational Event Prediction in Distribution Feeders

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Abstract—Distribution feeders with high photovoltaic penetration face increasing operational risks, including voltage violations, reverse power flow, and export events under time-varying demand and renewable uncertainty. This paper proposes a risk-aware graph learning framework that generates diverse power-network rollouts with contingency, derating, and overload scenarios, and represents each operating point as a graph snapshot. A multi-task graph neural network surrogate is developed to jointly perform node-level, edge-level, and graph-level prediction through a shared encoder with task-specific heads.

The model simultaneously learns voltage magnitude regression, node voltage violation classification, edge reverse power flow detection, and export event classification. Experiments are conducted on two power-network test systems, IEEE 33-bus and IEEE 118-bus, with repeated runs over three random seeds. On 33-bus, the main GraphSAGE model achieves 0.0511 ± 0.0010 voltage-magnitude RMSE, together with AUCs of 0.9178 ± 0.0002 , 0.9810 ± 0.0006 , and 0.9666 ± 0.0003 for node voltage violation, edge reverse power flow, and export prediction, respectively. On 118-bus, the corresponding results are 0.0067 ± 0.0003 RMSE and AUCs of 0.9885 ± 0.0001 , 0.9888 ± 0.0001 , and 0.8973 (std < 0.0001). Extensive studies, including backbone and feature ablations, graph-level classical baselines, cross-system comparison, and risk subtype analysis, highlight both the effectiveness and the limitations of the proposed approach. Overall, the framework provides a unified surrogate model for efficient monitoring, scenario screening, and multi-level operational risk prediction.

Index Terms—graph neural network, distribution feeder, photovoltaic integration, reverse power flow, voltage violation, multi-task learning, risk-aware modeling

I. INTRODUCTION

Distribution feeders are becoming increasingly dynamic because of high photovoltaic (PV) penetration, time-varying demand, and uncertain operating conditions, especially under increasing distributed generation integration in modern power systems [1]. Under such settings, feeder operators may face several important events, including node voltage violations, reverse power flow (RPF) on lines, and export toward the upstream grid. Fast and reliable prediction of these events is useful for feeder monitoring, screening of stressed operating points, and what-if analysis.

Conventional AC power flow remains the standard tool for accurate feeder analysis, but repeated simulation over long

horizons and many stressed scenarios can be computationally expensive. This has motivated growing interest in machine-learning-based approximations and predictive surrogates for power-system applications [2]. At the same time, data-driven surrogates are often trained for only one output type, such as bus voltage regression or graph-level status classification. In practice, however, feeder analysis is inherently multi-level: one may care about node-level voltage magnitude and voltage-violation status, edge-level reverse power flow events, and graph-level feeder export status simultaneously.

While existing works typically focus on isolated prediction tasks such as voltage estimation or feeder-level classification, practical feeder operation requires simultaneous awareness across multiple structural levels, including node conditions, line behaviors, and system-level events. Treating these tasks independently often leads to fragmented models and inconsistent predictions across levels. In this work, we instead formulate feeder monitoring as a unified multi-level prediction problem, where node-, edge-, and graph-level behaviors are learned jointly under diverse operating conditions. This perspective shifts the focus from optimizing individual tasks to building a coherent surrogate model that captures cross-level dependencies in distribution systems.

The main contributions of this work are as follows:

- We formulate feeder monitoring as a unified multi-level prediction problem and develop a multi-task graph neural surrogate to support node-, edge-, and graph-level inference within a single framework.
- We construct graph snapshots with explicit node features, edge features, and multi-level supervision, enabling unified learning across node, edge, and graph tasks.
- We design a multi-task graph neural network (GNN) surrogate that jointly predicts voltage magnitude, node voltage violations, edge reverse power flow, and export events.
- We present an empirical study across two systems, including repeated-seed evaluation, backbone ablation, edge-feature ablation, graph-level classical baselines, and risk subtype analysis.

Results show that the proposed GNN is particularly effective for structured node- and edge-level prediction tasks, while graph-level indicators remain competitive with simpler statistical baselines.

II. RELATED WORK

Graph neural networks (GNNs) have become a natural choice for power-system applications because electrical networks are intrinsically graph structured. Recent work has applied graph neural networks to infrastructure modeling, state prediction, and structured classification [3]–[7]. In parallel, broader machine learning literature has shown that multi-task learning can improve efficiency and representation quality when several related outputs share latent structure [8]. For power-system applications more generally, recent reviews also highlight growing interest in machine learning methods for system analysis and operation [2]. Voltage-related operating issues remain a long-standing concern in electric power systems [9].

Despite this progress, two gaps remain relevant here. First, many learning pipelines are trained mainly under nominal conditions and do not explicitly incorporate risk-driven operating changes such as contingency-like disturbances, line derating, or overload events. Second, many power-network surrogates are designed for only one structural level, for example bus regression or feeder-level classification, instead of learning node-, edge-, and graph-level targets together.

Our work addresses these limitations by combining risk-aware power-network simulation and multi-task graph learning. Specifically, we generate diverse risk-aware rollouts of-line, convert each solved operating point into a graph snapshot, and train a shared graph encoder with task-specific heads for node-, edge-, and graph-level prediction. This unified formulation enables simultaneous learning across multiple structural levels while explicitly capturing risk-relevant operating conditions.

III. RISK-AWARE GRAPH LEARNING FRAMEWORK

A. Overview

Fig. 1 summarizes the four-stage workflow: risk-aware rollout generation, graph snapshot construction, multi-task GNN learning, and evaluation. Risk injection exposes the model to stressed regimes beyond nominal operation and enriches the operating scenarios represented in the dataset.

B. Risk-Aware Rollout Generation

We build the offline dataset from a pandapower-based feeder simulation backend [10]. In the current study, the framework is evaluated on both the `case33bw` 33-bus distribution network and an IEEE 118-bus power-network setting. PV units are placed on buses selected by a farthest-from-slack strategy, with random PV capacities sampled within predefined ranges. Time-varying load and PV profiles are generated from daily scaling factors, intra-day temporal shapes, cloud-driven PV fluctuations, and forecast perturbations.

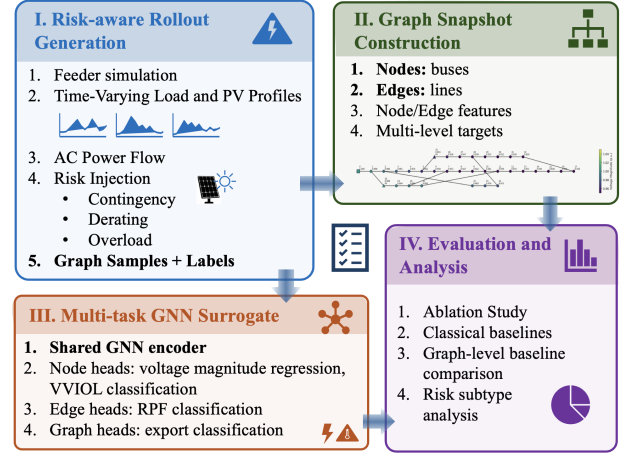


Fig. 1: Overview of the proposed framework.

At each time step, the backend updates load and PV injections and runs AC power flow to obtain the system operating state. The resulting solved states are then used to construct graph snapshots for learning and evaluation.

To model stressed operating conditions, we introduce three operational risk subtypes:

- **Contingency:** temporary out-of-service events on selected elements, implemented primarily on lines;
- **Derating:** temporary reduction of line current capability through a random derating factor;
- **Overload:** temporary load amplification through a multiplicative stress factor.

These events are managed by a risk-state controller with per-step trigger probabilities and finite durations. Only valid solved operating points are retained as training samples.

C. Graph Snapshot Construction

Each valid solved system state is converted into a graph snapshot. Buses are treated as nodes and lines are treated as edges. Let the graph at time step t be denoted by

$$G_t = (V_t, E_t, X_t, Z_t), \quad (1)$$

where X_t is the node feature matrix and Z_t is the edge feature matrix.

In our implementation, each bus i is represented by an 8-dimensional node feature vector

$$\mathbf{x}_{i,t} = \left[\mathbf{1}_{\{i=\text{slack}\}}, \frac{v_i^{\text{base}}}{V_{\text{base}}}, P_i^{\text{base}}, \hat{P}_{i,t}^L, \hat{Q}_{i,t}^L, \hat{P}_{i,t}^{PV}, \hat{Q}_{i,t}^{PV}, \tilde{v}_{i,t-1} \right], \quad (2)$$

where $\mathbf{1}_{\{i=\text{slack}\}}$ is a slack-bus indicator, v_i^{base} is the nominal bus voltage, V_{base} is a fixed voltage normalization constant, P_i^{base} is the base active load at bus i , $\hat{P}_{i,t}^L$ and $\hat{Q}_{i,t}^L$ are the forecast active and reactive load, $\hat{P}_{i,t}^{PV}$ and $\hat{Q}_{i,t}^{PV}$ are the forecast PV injections, and $\tilde{v}_{i,t-1}$ is the previous-step voltage observation with measurement noise. The forecast quantities are generated within the rollout engine: active-load and PV forecasts are obtained by perturbing profile multipliers with Gaussian noise, reactive-load forecast is derived from base reactive demand or a fixed reactive-to-active ratio, and forecast

PV reactive injection is set to zero. The final node feature matrix is

$$X_t = \begin{bmatrix} \mathbf{x}_{1,t}^\top \\ \mathbf{x}_{2,t}^\top \\ \vdots \\ \mathbf{x}_{|V|,t}^\top \end{bmatrix} \in \mathbb{R}^{|V| \times 8}. \quad (3)$$

Each physical line is converted into two directed edges in the graph. For each directed edge, we use a 2-dimensional static feature vector

$$\mathbf{z}_{ij} = [r_{ij}\ell_{ij}, x_{ij}\ell_{ij}], \quad (4)$$

where r_{ij} and x_{ij} denote the per-kilometer resistance and reactance of the corresponding physical line and ℓ_{ij} is the line length. This yields the total series resistance and total series reactance of the line segment. If the physical network contains M undirected lines, the graph representation uses $|E| = 2M$ directed edges, and the edge feature matrix satisfies

$$Z_t \in \mathbb{R}^{|E| \times 2}. \quad (5)$$

The supervision includes four targets across three structural levels: node-level voltage magnitude regression and voltage-violation classification, edge-level reverse-power-flow classification, and graph-level export classification.

The continuous node-level target is the solved voltage magnitude

$$y_{i,t}^{\text{vm}} = v_{i,t}. \quad (6)$$

The three binary event labels are defined as

$$y_{i,t}^{\text{vviol}} = \mathbf{1}_{\{v_{i,t} < v_{\min} \vee v_{i,t} > v_{\max}\}}, \quad (7)$$

$$y_{e,t}^{\text{rpf}} = \mathbf{1}_{\{p_{e,t} < 0\}}, \quad (8)$$

$$y_t^{\text{export}} = \mathbf{1}_{\{p_t^{\text{ext}} < 0\}}, \quad (9)$$

where $v_{i,t}$ is the solved voltage magnitude at bus i , $p_{e,t}$ is the directed active power flow on edge e , and p_t^{ext} is the active power at the external grid. Accordingly, the three binary labels indicate voltage violation, reverse power flow, and feeder export.

D. Multi-Task GNN Surrogate

The model uses one shared graph encoder and four task-specific heads. This design follows the general motivation of multi-task learning, where related prediction objectives benefit from a shared representation while retaining task-specific output mappings [8]. The encoder backbone can be instantiated as MLP, GCN, GAT, or GraphSAGE. GraphSAGE is adopted as the primary backbone in the main experiments, while the other variants are evaluated in the ablation study. Given a graph snapshot $G_t = (V_t, E_t, X_t, Z_t)$, the shared encoder produces node embeddings

$$H_t = f_{\text{enc}}(X_t, E_t, Z_t) \in \mathbb{R}^{|V| \times d}, \quad (10)$$

where d is the hidden dimension. A graph-level embedding $\mathbf{g}_t$ is then obtained by global mean pooling over the node embeddings,

$$\mathbf{g}_t = \text{MeanPool}(H_t) = \frac{1}{|V_t|} \sum_{i \in V_t} \mathbf{h}_{i,t}, \quad (11)$$

where $\mathbf{h}_{i,t}$ denotes the embedding of node i at time step t . The graph embedding $\mathbf{g}_t$ is used by the export head for graph-level prediction.

Task-specific heads operate on these shared representations:

$$\hat{\mathbf{v}}_t = f_{\text{vm}}(H_t), \quad (12)$$

$$\hat{\mathbf{s}}_t^{\text{vviol}} = f_{\text{vviol}}(H_t), \quad (13)$$

$$\hat{\mathbf{s}}_t^{\text{rpf}} = f_{\text{rpf}}(H_t, E_t, Z_t), \quad (14)$$

$$\hat{\mathbf{s}}_t^{\text{export}} = f_{\text{export}}(\mathbf{g}_t), \quad (15)$$

where $\hat{\mathbf{v}}_t$ is the node-level voltage regression output and $\hat{\mathbf{s}}^{(\cdot)}$ denote logits for the classification tasks.

The overall training objective is

$$\mathcal{L} = \lambda_1 \mathcal{L}_{\text{vm}} + \lambda_2 \mathcal{L}_{\text{vviol}} + \lambda_3 \mathcal{L}_{\text{rpf}} + \lambda_4 \mathcal{L}_{\text{export}}, \quad (16)$$

where the voltage regression term is defined as

$$\mathcal{L}_{\text{vm}} = \frac{1}{|V|} \sum_{i \in V} (\hat{v}_{i,t} - v_{i,t})^2, \quad (17)$$

and the three event-prediction losses are binary cross-entropy losses with logits:

$$\mathcal{L}_{\text{cls}} = -\frac{1}{N} \sum_{n=1}^N [w^+ y_n \log \sigma(s_n) + (1 - y_n) \log(1 - \sigma(s_n))], \quad (18)$$

where $\sigma(\cdot)$ is the sigmoid function and w^+ denotes the positive-class weight used to address class imbalance. Equation (18) is instantiated separately for node voltage violation, edge reverse power flow, and graph export classification. Positive-class weighting is enabled by default for the classification heads, and decision thresholds are selected on the validation set rather than fixed at 0.5. Node voltage-violation classification is more challenging than voltage-magnitude regression because it depends on hard thresholding of the continuous voltage prediction around fixed operating limits ($v_{\min} = 0.95$ p.u. and $v_{\max} = 1.05$ p.u.), so even small regression errors near the boundary may flip the event label.

E. Graph-Level Aggregation for Evaluation

In addition to the native export head, we evaluate graph-level event prediction by aggregating node- and edge-level probabilities. Specifically,

$$s_{\text{any_vviol}}(G_t) = \max_{i \in V_t} p_{i,t}^{\text{vviol}}, \quad (19)$$

$$s_{\text{any_rpf}}(G_t) = \max_{e \in E_t} p_{e,t}^{\text{rpf}}, \quad (20)$$

with graph-level binary labels

$$y_{\text{any_vviol}}(G_t) = \mathbf{1}_{\{\sum_{i \in V_t} y_{i,t}^{\text{vviol}} > 0\}}, \quad (21)$$

$$y_{\text{any_rpf}}(G_t) = \mathbf{1}_{\{\sum_{e \in E_t} y_{e,t}^{\text{rpf}} > 0\}}. \quad (22)$$

A graph is positive for *any-VVIOL* if any node is in violation and positive for *any-RPF* if any directed edge exhibits reverse power flow. This evaluation is stricter than the native graph-head setting because it tests whether localized predictions can be meaningfully aggregated.

IV. EXPERIMENTAL SETUP

A. Dataset and Split

We evaluate the framework on two simulated power-network datasets. For the 33-bus system, the dataset contains 7263 valid graph snapshots after filtering invalid power-flow steps, with train/validation/test splits of 5810/726/727. For the 118-bus system, the corresponding splits are 7641/960/960. The two systems provide different subtype distributions for risk-aware evaluation, with the 118-bus setting offering a larger subtype breakdown.

TABLE I: Dataset summary for the two systems.

Item	33-bus	118-bus
Train / Val / Test	5810 / 726 / 727	7641 / 960 / 960
Node feature dim.	8	8
Edge feature dim.	2	2
Test no-risk	586	431
Test contingency	8	313
Test derating	66	117
Test overload	67	99

For 33-bus, the contingency subset remains small ($n = 8$ in the test set), so subtype-specific results on this system are interpreted cautiously. The 118-bus risk subtype analysis provides a complementary and more stable view of regime-dependent behavior.

B. Model, Training, and Thresholding

The model backbones are compared across GraphSAGE, GCN, GAT, and MLP. The loss function weights are set as follows: $\lambda_{vm} = 1.0$, $\lambda_{vviol} = 1.0$, $\lambda_{rpf} = 1.0$, and $\lambda_{export} = 0.5$.

For threshold tuning, a grid over $[0.01, 0.99]$ is scanned on the validation set. The node-voltage-violation and edge-RPF thresholds are selected by validation F1, and the export threshold by validation accuracy. These are decision thresholds on sigmoid outputs, distinct from the physical voltage limits defining voltage violations.

To improve statistical reliability, the main GraphSAGE results are reported over three random seeds on both 33-bus and 118-bus. We summarize performance using mean and standard deviation.

C. Implementation Details

All models are implemented in PyTorch and PyTorch Geometric. For the main results, we use a 3-layer GraphSAGE encoder with hidden dimension 128, dropout 0.3, batch size 32, gradient clipping 1.0, and validation-based model selection over 30 training epochs. For repeated-seed experiments, we use seeds 12346, 12347, and 12348, and report mean and standard deviation over the resulting test metrics.

D. Classical Graph-Level Baselines

To compare against non-graph models at the graph level, we construct graph-summary feature vectors by aggregating node and edge attributes through mean, standard deviation, minimum, and maximum statistics. We additionally include graph size features and, in the strongest baseline configuration reported here, explicit risk metadata (`risk_active`, `risk_type`). Logistic regression and random forest are trained on these graph-level summaries. A weak constant-score rule baseline is also included for reference.

This comparison should be interpreted carefully: the classical baselines are trained directly on graph-level handcrafted summaries, while the GNN is trained jointly on node-, edge-, and graph-level tasks and only later aggregated for *any-VVIOL* and *any-RPF*. Thus, the baseline comparison evaluates both modeling power and task formulation.

E. Evaluation Metrics

We report RMSE and MAE for voltage regression, AUC and F1 for node and edge event classification, and AUC together with accuracy for export prediction.

V. RESULTS

A. Main Multi-Task Results

Table II reports the main multi-task GraphSAGE results on the two systems, summarized over three random seeds. The model performs strongly across all four tasks on both systems, showing that one shared encoder can support both regression and classification across multiple structural levels.

The strongest structured prediction results are observed for edge RPF on both systems. The two systems also exhibit different difficulty profiles: the 118-bus system is easier for voltage regression and edge-level classification, while export prediction is stronger on 33-bus.

These cross-system results clarify the scope of the proposed surrogate. Rather than suggesting that one system is uniformly easier than the other, the results indicate that task difficulty depends on both network structure and event definition. In particular, the 118-bus system shows stronger voltage-regression and edge-level discrimination, while export prediction is relatively more favorable on 33-bus.

B. Backbone and Training Ablations

Fig. 3 compares four encoder backbones under the same multi-task setting. Message-passing backbones consistently outperform the MLP baseline on structured network tasks, especially for edge RPF prediction, where line interactions matter directly. Among them, GraphSAGE provides the strongest overall balance.

Table III reports selected ablation results on the 118-bus system relative to the main GraphSAGE model. Several points are notable. First, removing edge attributes causes only minor degradation in the present setting, suggesting that much of the predictive signal is already available from topology and node-side state context. Second, disabling positive-class weighting has mixed effects but does not improve the overall balance

TABLE II: Main multi-task test performance of the proposed GraphSAGE surrogate (mean $\pm$ std over 3 seeds).

System	VM RMSE	VVIOL AUC / F1	RPF AUC / F1	Export AUC / ACC
33-bus	0.0511 ± 0.0010	0.9178 ± 0.0002 / 0.7530 ± 0.0015	0.9810 ± 0.0006 / 0.8280 ± 0.0031	0.9666 ± 0.0003 / 0.9158 ± 0.0026
118-bus	0.0067 ± 0.0003	0.9885 ± 0.0001 / 0.7162 ± 0.0004	0.9888 ± 0.0001 / 0.9472 ± 0.0012	0.8973 (std < 0.0001) / 0.8240 (std = 0)

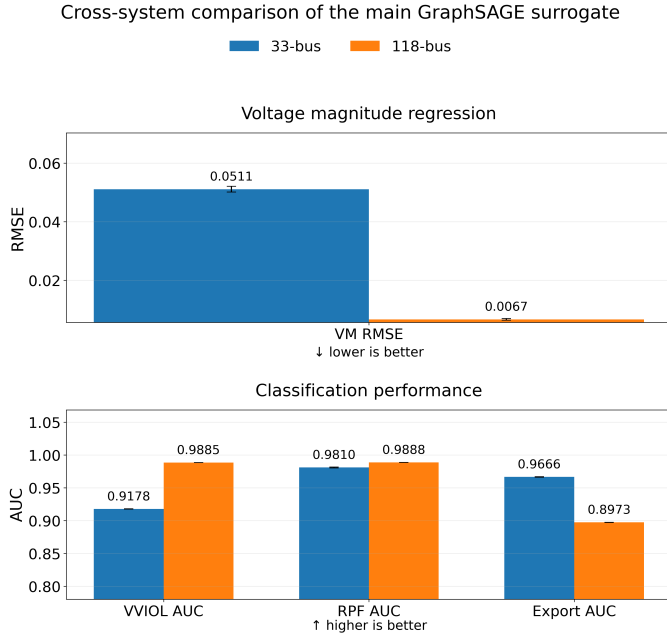


Fig. 2: Cross-system comparison of the main GraphSAGE surrogate on 33-bus and 118-bus. The two systems show different task difficulty profiles, motivating evaluation beyond a single system.

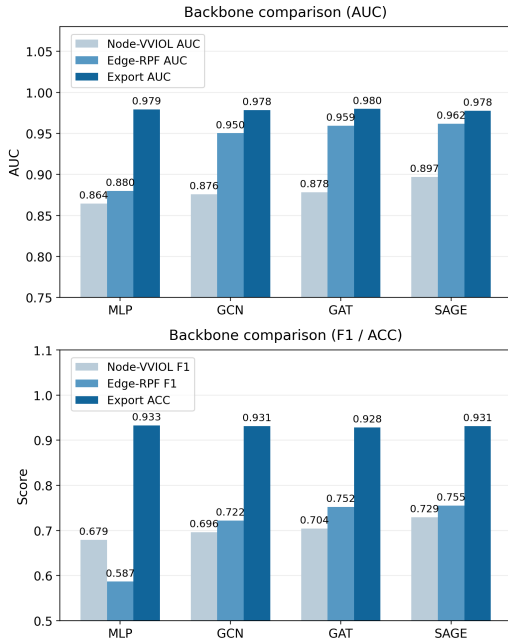


Fig. 3: Backbone comparison under the same multi-task setting. Message-passing backbones outperform MLP, while GraphSAGE provides the strongest overall balance.

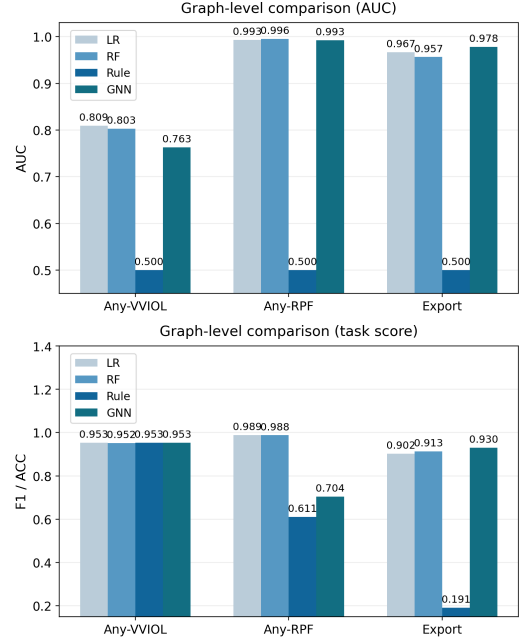


Fig. 4: Graph-level comparison on 33-bus under the same task. The proposed GNN excels at export, while classical baselines remain competitive on some aggregated indicators.

across tasks. Third, using a fixed threshold of 0.5 substantially hurts node-VVIOL F1, confirming the importance of validation-tuned thresholds.

The same qualitative trends are also observed on 33-bus: MLP is weaker than message-passing backbones, and fixed thresholds are consistently less reliable than validation-tuned thresholds.

C. Graph-Level Comparison With Classical Baselines

Table IV and Fig. 4 compare graph-level GNN outputs with classical baselines on 33-bus. The graph-level GNN obtains AUC values of 0.7629 for *any-VVIOL*, 0.9928 for *any-RPF*, and 0.9783 for export. However, the strongest classical baselines remain highly competitive on these aggregated tasks: logistic regression reaches AUC = 0.7933 on *any-VVIOL*, while random forest reaches AUC = 0.9955 on *any-RPF*.

These results highlight task-dependent strengths rather than a simple winner-take-all comparison. Classical baselines are well matched to coarse graph-level labels because they are trained directly on handcrafted graph summaries, whereas the proposed GNN is optimized jointly for node-, edge-, and graph-level prediction. Its main advantage lies in unified structured prediction rather than dominance on every aggregated graph statistic.

This observation also helps explain why classical graph-level baselines remain competitive on some aggregated tasks.

TABLE III: Ablation results on 118-bus reported as differences (Δ) relative to the main GraphSAGE multi-task model. Positive Δ in RMSE indicates worse regression; negative Δ in AUC/F1/ACC indicates worse classification.

Setting	Δ RMSE	Δ VVIOL AUC	Δ VVIOL F1	Δ RPF AUC	Δ RPF F1	Δ Export AUC	Δ Export ACC
Main GraphSAGE (Ref.)	0.0067	0.9885	0.7162	0.9888	0.9472	0.8973	0.8240
MLP backbone	+0.0114	-0.0053	-0.0001	-0.0052	-0.0159	-0.0002	-0.0021
GCN backbone	+0.0037	-0.0003	-0.0009	-0.0020	-0.0060	+0.0001	-0.0010
GAT backbone	+0.0007	+0.0008	-0.0042	-0.0005	-0.0036	+0.0000	+0.0000
No edge attributes	+0.0003	+0.0005	-0.0013	+0.0000	-0.0007	+0.0000	-0.0021
No positive weighting	-0.0004	+0.0006	+0.0004	+0.0004	+0.0021	-0.0003	-0.0010
Fixed threshold 0.5	+0.0033	+0.0001	-0.3062	-0.0007	-0.0035	-0.0001	-0.0010

TABLE IV: Graph-level comparison on 33-bus.

Task	LR	RF	MLP	GNN
Any-VVIOL (AUC)	0.7933	0.7767	0.7590	0.7629
Any-VVIOL (F1)	0.9532	0.9491	0.9532	0.9532
Any-RPF (AUC)	0.9933	0.9955	0.9938	0.9928
Any-RPF (F1)	0.9892	0.9861	0.9892	0.7041
Export (AUC)	0.9671	0.9566	0.9665	0.9783
Export (ACC)	0.9161	0.8913	0.9216	0.9298

Once node- and edge-level states are compressed into coarse graph summaries, a portion of the structural advantage of message passing is reduced, especially for labels such as any-VVIOL and any-RPF that depend on max-style aggregation. In contrast, the proposed GNN is designed to preserve fine-grained node- and edge-level information and to support unified prediction across structural levels. Its value therefore lies less in dominating every graph-level baseline and more in providing a consistent multi-level surrogate within one model.

D. Risk Subtype Analysis

Risk subtype analysis reveals that performance depends strongly on operating regime, and that these regime-dependent effects differ across systems.

Table V summarizes key subtype-specific metrics for both systems. On 33-bus, overload and derating produce clearer node-level stress signatures. On 118-bus, the larger subtype partitions enable a more stable subtype-wise comparison. In particular, no-risk cases are easiest for node-VVIOL and RPF, while overload produces the weakest export accuracy and the worst voltage-regression error. These results show that one average test number can obscure substantial regime-dependent variation.

VI. DISCUSSION

The experiments suggest four main observations.

First, multi-task graph learning is effective across node-, edge-, and graph-level outputs. One shared encoder can support structurally different tasks without requiring separate task-specific models.

Second, explicit message passing matters. MLP is consistently weaker than GCN, GAT, and GraphSAGE, especially on edge reverse power flow prediction, where line interactions and graph structure are directly relevant.

Third, the main advantage of the proposed model is unified structured prediction rather than superiority on every

aggregated graph-level label. This is particularly relevant for monitoring settings in which multiple operational indicators must be assessed jointly rather than through separate task-specific models.

Fourth, evaluating on more than one system materially changes the strength of the empirical evidence. The 33-bus and 118-bus systems exhibit different difficulty profiles, and the 118-bus system provides a more stable risk subtype analysis because its subtype partitions are larger. This additional system-level evidence helps mitigate the limitation of drawing conclusions from only one network topology.

This study still has several limitations. First, although evaluation now spans two systems, broader cross-topology generalization remains to be validated on a larger family of networks. Second, the risk scenarios are simulation-generated rather than derived from field measurements. These limitations motivate future work on larger multi-network evaluation, richer real-world operating data, broader risk modeling, and tighter integration with control-oriented applications.

Beyond prediction, the proposed framework can also support learning-based operational decision support. In practical feeder operation, predictions are often coupled with actions such as reconfiguration, curtailment, or setpoint adjustment. A unified multi-task surrogate therefore provides a natural foundation for future decision-oriented pipelines.

VII. CONCLUSION

This paper presented a risk-aware multi-task GNN surrogate for operational event prediction in distribution feeders. The framework combines risk-aware rollout generation, graph snapshot construction, and joint learning across node-, edge-, and graph-level tasks. Experiments on both 33-bus and 118-bus systems show strong performance on voltage regression, voltage-violation detection, reverse-power-flow detection, and export prediction, together with informative ablation, baseline, and subtype analyses.

More importantly, the added cross-system and repeated-seed results strengthen the empirical reliability of the framework beyond a single-system study. The 118-bus results provide a more stable complementary view of regime-dependent performance. Overall, the proposed surrogate serves as a compact and efficient screening model for stressed feeder operating conditions.

Future work will extend the framework to additional network topologies, richer edge and temporal representations,

Risk-type performance comparison on 33-bus

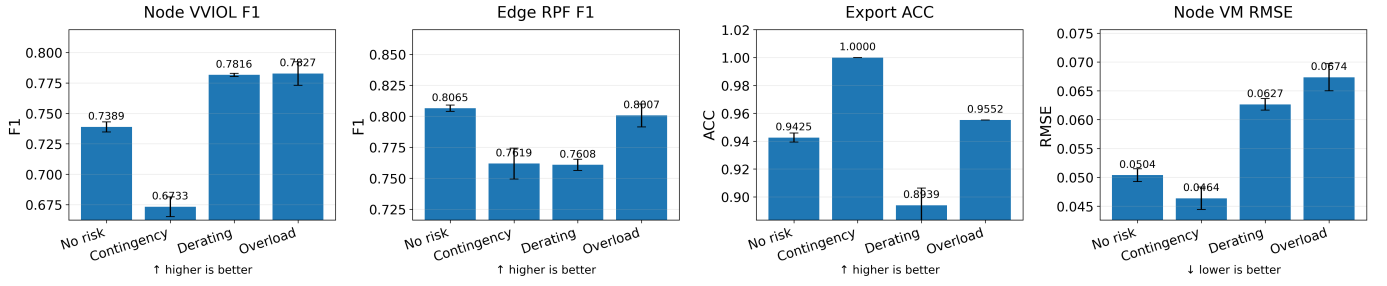


Fig. 5: Risk-type performance comparison on 33-bus.

Risk-type performance comparison on 118-bus

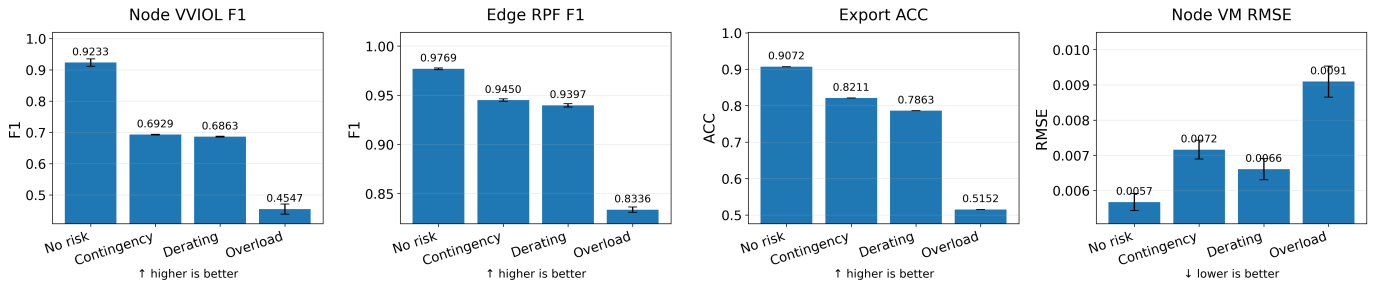


Fig. 6: Risk-type performance comparison on 118-bus. Compared with 33-bus, the larger subtype partitions provide a more stable view of regime-dependent behavior.

TABLE V: Risk subtype analysis of the main GraphSAGE model (mean over 3 seeds).

Subtype	33-bus				118-bus			
	VVIOL F1	RPF F1	Export ACC	VM RMSE	VVIOL F1	RPF F1	Export ACC	VM RMSE
No risk	0.7389	0.8065	0.9425	0.0504	0.9233	0.9769	0.9072	0.0057
Contingency	0.6733	0.7619	1.0000	0.0464	0.6929	0.9450	0.8211	0.0072
Derating	0.7816	0.7608	0.8939	0.0627	0.6863	0.9397	0.7863	0.0066
Overload	0.7827	0.8007	0.9552	0.0674	0.4547	0.8336	0.5152	0.0091

improved graph-level aggregation, and integration with control and optimization pipelines for decision-aware learning.

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